Design & evaluation of MNIST neural network

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MSc Data Science
Machine Learning
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Aim and objectives

- Develop an artificial neural network (ANN) to classify handwritten digits from 0 to 9
- Prepare dataset and create a training and validation set
- Evaluate performance using accuracy, loss and confusion matrix
- Compare models with and without dropout regularisation
- Assess results, limitations and potential model improvements

MNIST dataset

- 70,000 images of handwritten digits in total
- Each image is 28x28 pixels and greyscale
- Ten possible classes: digits 0-9
- 60,000 images supplied as training data, 10,000 reserved as test data
- Supplied training data split into 54,000 training images and 6,000 validation images

Dataset Partitioning

54,000 Training images

6,000 Validation images

10,000 Test images

- Training set: teaches the neural network to recognise patterns
- Validation set: helps assess performance during development
- Test set: provides a final assessment using separate images
- A stratified split was used to maintain similar proportions of each digit class

Data preprocessing

- Images presented as 28x28 grey scale pixel arrays
- Pixel values originally ranged from 0 to 255
- Values were normalised by dividing by 255, giving range to 0 to 1
- Training, validation and test images received the same normalisation
- original digit labels (0-9) were retained

Designing the ANN

Input image
28x28 pixels

Flatten
Converts image into 784 values

Dense layer:128 neurons
ReLU activation

Dropout: 20%
Regularisation during training

Dense layer: 64 neurons
ReLU activation

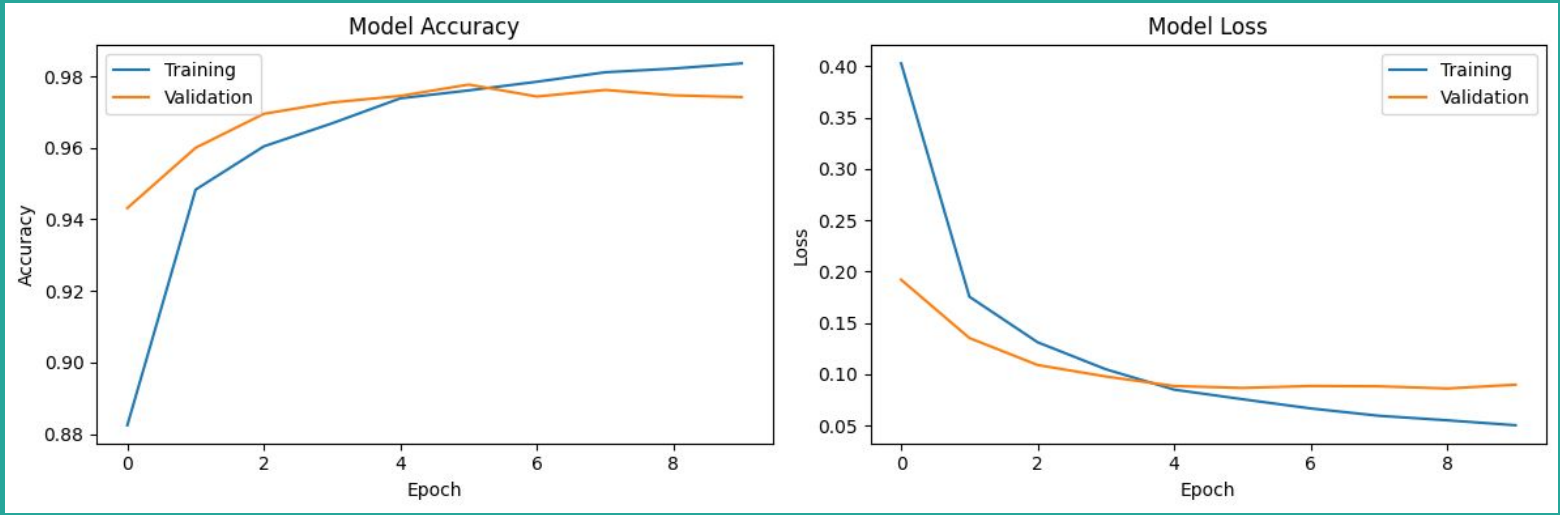
Output layer: 10 neurons
Softmax- probabilities for digits 0-9

Training configuration

- Optimiser: Adam
- Loss function: sparse categorical cross entropy
- Batch size: 128
- Training duration: 10 epochs

Training Results

Training Performance Across 10 Epochs



- Training accuracy increased as the model learned
- validation accuracy reached a best recorded value of 97.8%
- Difference between training and validation performance suggests some overfitting may be present

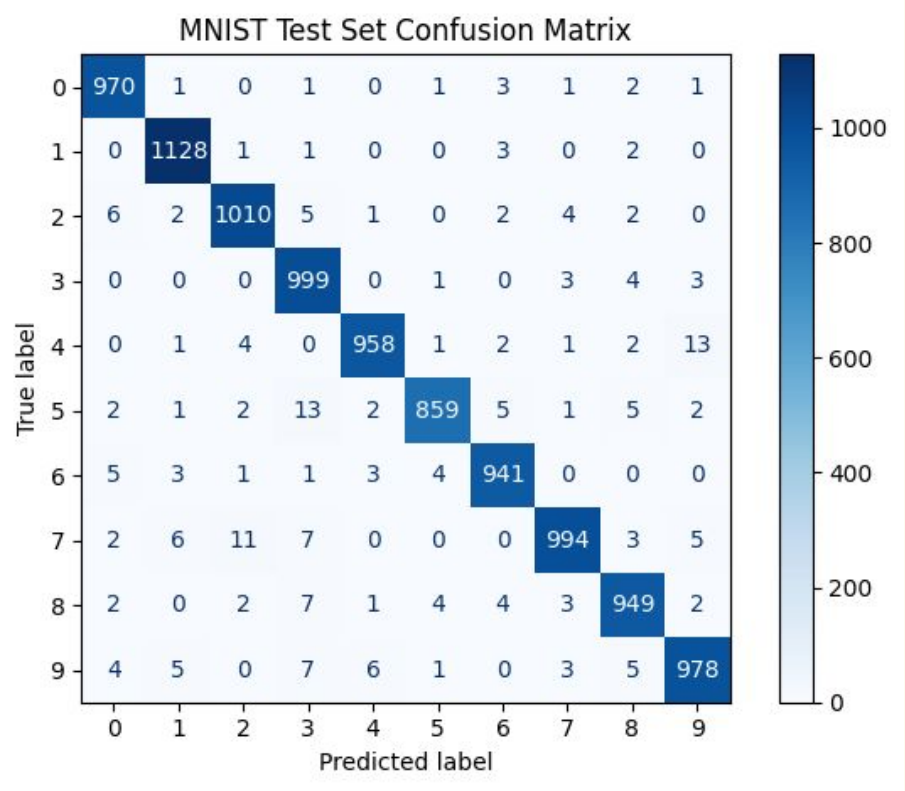
Final Model Performance

Test accuracy

97.86%

Correct predictions

9,786 out of 10,000 images



- 214 images were misclassified
- Model performed well overall
- Some digit pairs were more frequently confused than others

Comparing models with and without dropout

	With dropout	Without dropout
Final training accuracy	98.36%	99.43%
Final validation accuracy	97.42%	97.45%
Validation loss	0.0899	0.0933
Epochs	10	10

Evaluation

- Achieved 97.86% test accuracy on MNIST
- Weakness: training and validation metrics show performance gap
- Design limitation: fully connected network flattens the images, losing their explicit two dimensional structure
- Dataset limitation: MNIST contains standardised, low resolution images and may not represent all real world handwriting
- Improvements: compare a convolutional neural network, repeat experiments with different random seeds and investigate misclassified images

Conclusion

- Developed and evaluated a neural network to classify handwritten digits
- Achieved 97.86% test accuracy
- Identified common classification errors using a confusion matrix
- Compared dropout and no-dropout configurations
- Found that high accuracy does not remove the need to assess overfitting, loss and generalisation

References

References

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